

CHƯƠNG 10

Các Xu hướng Dữ liệu và Phân tích Mới nhất trong Kế toán

Trí tuệ Nhân tạo cho Kế toán (AI in Accounting)

Khoa Kế toán - Kiểm toán

Bài giảng Thực hành 10

Intro 10.0

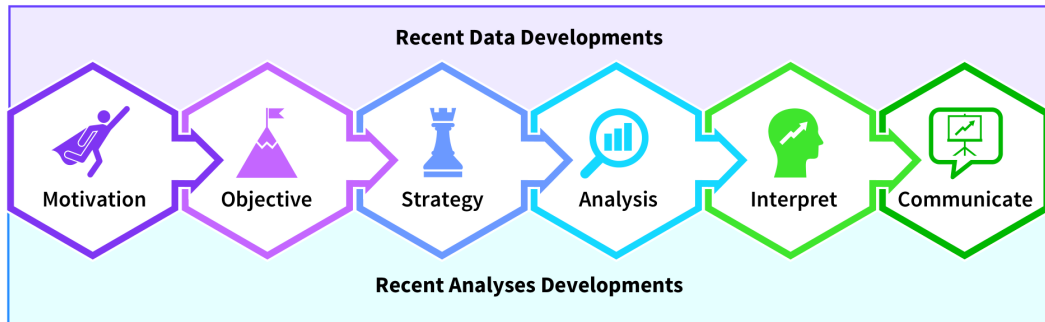
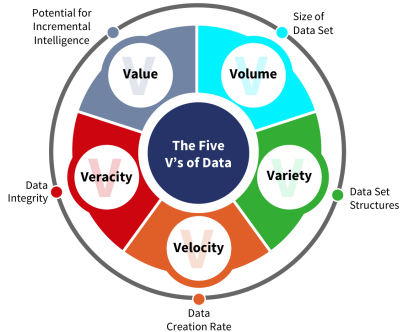
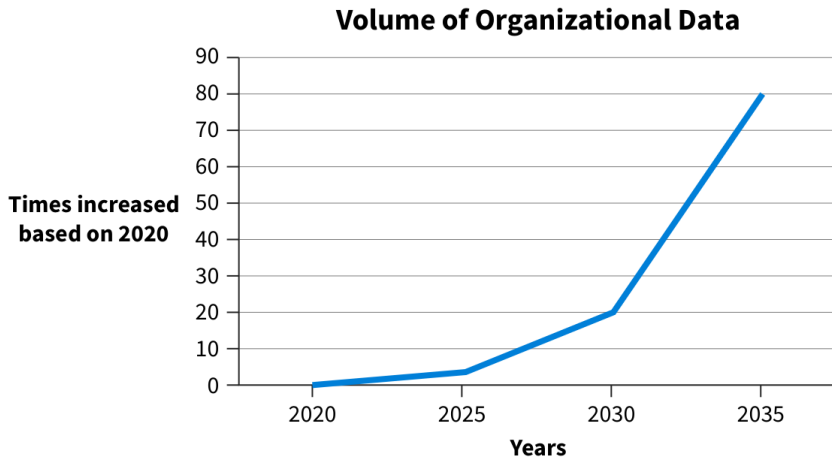


ILLUSTRATION 10.1



Characteristic	Examples
Volume	In AIS: • Process flow logs • Security alerts • System error logs • Transactions • Firewall traffic • System and application access logs
Variety	In auditing: • Codes, approval stamps, text, pictures, dates, and recorded values • Fair market values at year end
Velocity	In financial accounting: • Cash transactions • Stock price changes • Social media content
Veracity	In managerial accounting: • Transactions • Prices and costs • Vendor and customer data • Sensor data from smart machines
Value	In tax accounting: • Local, national, and international tax changes • Sales tax jurisdictions and rates

ILLUSTRATION 10.2



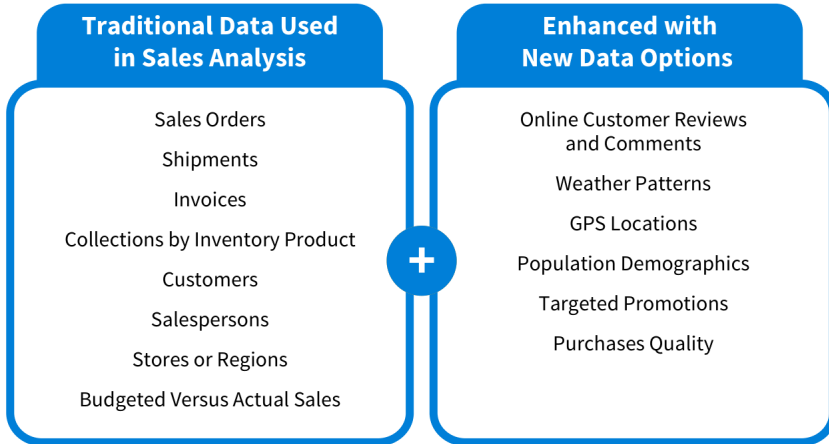
Chương 10: Các Xu hướng Dữ liệu và Phân tích Mới nhất

- **Mục tiêu:** Khám phá tác động của công nghệ và phân tích dữ liệu tiên tiến đối với nghề kế toán.
- **Trọng tâm:** Hiểu các đặc tính của Big Data và sự chuyển dịch của các công nghệ lõi.
- **Cấu trúc:** Dữ liệu lớn (Big Data), Sự phát triển của công nghệ, Ứng dụng vào các lĩnh vực kế toán.

ILLUSTRATION 10.3

Data Cloud	Benefits	Costs
Private Clouds		
Data storage for a single organization.	• Offers high data privacy and data security. • Offers high data interactivity. • More customization and adaptability are possible.	• Can be the most expensive option. • May be difficult to manage if developed in-house without hiring staff with cloud expertise.
Data storage that is shared by a set group of non-competing organizations.	• All the benefits of single organization storage, especially for data interactivity between supply chain partners. • Sharing data storage makes it a less expensive option for each company.	• Can have some restrictions on customization and adaptability. • May involve shared contracts and joint revisions.
Public Clouds		
Public clouds are shared by many organizations who can easily come and go from the public cloud.	• Lowest individual company cost. • Fastest implementation. • Most appropriate for small businesses.	Least adaptable or customizable cloud data service.
Hybrid Clouds		
Include both private and public cloud virtual server sections.	The benefits and costs associated with both private and public clouds also apply to hybrid clouds. Hybrid clouds allow organizations to choose which data subset is best stored in a private cloud, while storing the rest in a public cloud for the cost efficiencies.	

ILLUSTRATION 10.4



10.1 Đặc điểm của Dữ liệu lớn (5V của Big Data)

- Sự bùng nổ dữ liệu yêu cầu thay đổi trong phương pháp phân tích.
- Mô hình 5V giải thích bản chất đa chiều của Dữ liệu lớn:
 - Volume (Khối lượng)
 - Variety (Đa dạng)
 - Velocity (Tốc độ)
 - Veracity (Tính xác thực)
 - Value (Giá trị)

ILLUSTRATION 10.5

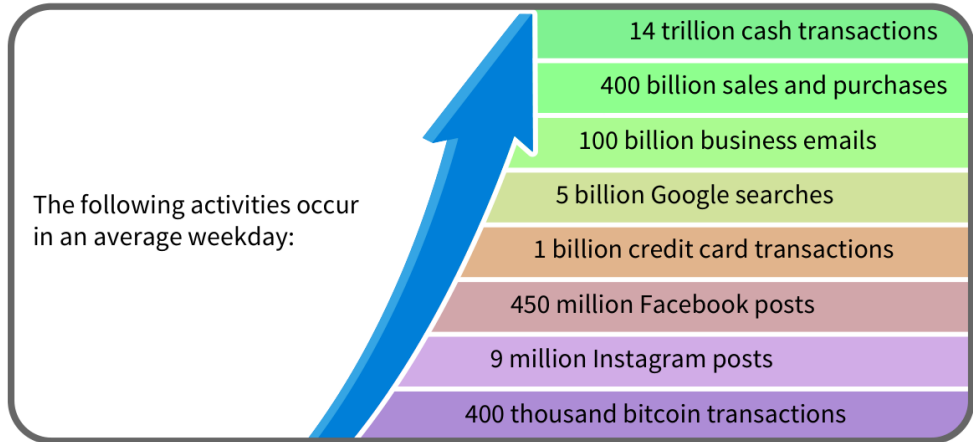


ILLUSTRATION 10.6

	Income Statement			
	Product A	Product B	Both Product Lines	Only Product A
Sales Revenues	\$720,000	\$1,200,000	\$1,920,000	720,000
less Variable Costs	432,000	960,000	1,392,000	432,000
= Contribution Margin	288,000	240,000	528,000	288,000
– Avoidable Fixed Expenses	210,000	225,000	435,000	210,000
= Segment Margin	78,000	15,000	93,000	78,000
– Unavoidable Common Fixed Expenses	42,000	42,000	84,000	84,000
= Net Income	<u><u>\$ 36,000</u></u>	<u><u>\$ (27,000)</u></u>	<u><u>\$ 9,000</u></u>	<u><u>\$ (6,000)</u></u>

Volume (Khối lượng) & Variety (Đa dạng)

- **Volume:** Lượng dữ liệu vượt ra khỏi khả năng xử lý của Excel hoặc SQL thông thường. Được tính bằng Terabyte, Petabyte,...
- **Variety:**
 - Dữ liệu có cấu trúc (Bảng tính, CSDL)
 - Dữ liệu bán cấu trúc (XML, JSON, XBRL)
 - Dữ liệu phi cấu trúc (Văn bản, Hình ảnh, Âm thanh) chiếm đến 80% tổng lượng dữ liệu hiện nay.

ILLUSTRATION 10.7

Law	Description
The Privacy Act of 1974	Restricts use of data held by government agencies.
Health Information Privacy Protection Act of 2013 and the Health Insurance Portability and Accountability Act of 1996 (HIPAA)	Restricts the sharing of individual's health care data, with fines up to \$10,000 per violation.
The Fair Credit Reporting Act (FCRA)	Limits and protects the credit-related personal data collected and shared.
The Family Educational Rights and Privacy Act (FERPA)	Protects the privacy of educational records.
The Gramm-Leach Bliley Act (GLBA)	Requires loan and investment service providers to disclose personal data sharing.
The Electronic Communications Privacy Act (ECPA) and Patriot Acts	Restricts when and how individuals' private data communications can be legally surveilled.
Children's Online Privacy Protection Act (COPPA)	Restricts personal data collected from minors and extends to third parties' data use.

ILLUSTRATION 10.8



Velocity (Tốc độ) & Veracity (Tính xác thực)

- **Velocity:** Tốc độ sinh ra và luân chuyển của dữ liệu. Doanh nghiệp cần xử lý dữ liệu theo thời gian thực (Real-time).
- **Veracity:** Mức độ đáng tin cậy. Dữ liệu từ mạng xã hội có thể bị nhiễu và sai lệch (bias), dẫn đến rủi ro ra quyết định sai lầm.

ILLUSTRATION 10.9

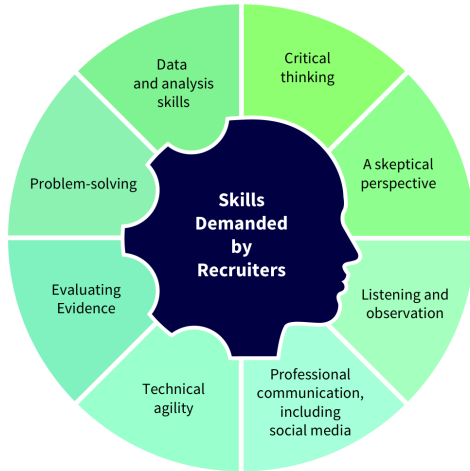
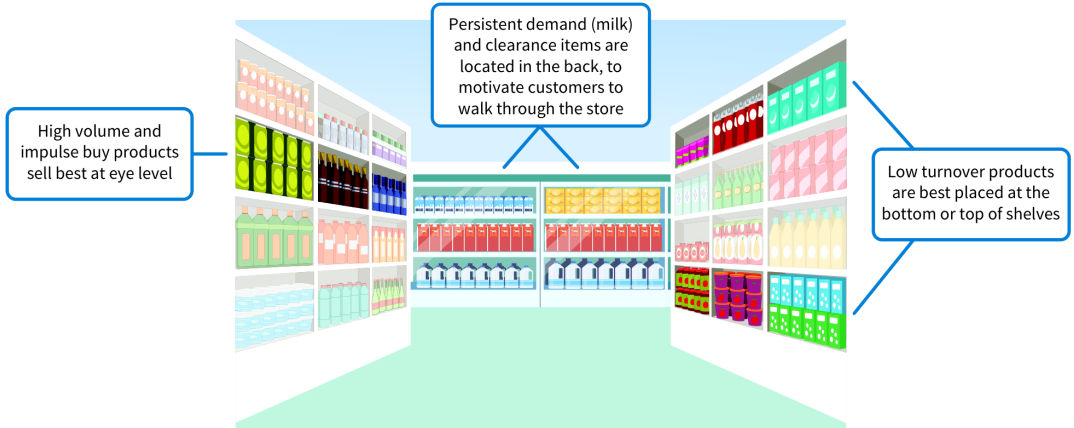


ILLUSTRATION 10.10



Value (Giá trị)

- **Value:** Tất cả 4V trước đều vô nghĩa nếu không thể trích xuất ra insight có giá trị (Value) cho doanh nghiệp.
- Đối với kế toán, Value thể hiện ở khả năng:
 - Dự báo dòng tiền chính xác hơn.
 - Nhận diện gian lận nhanh hơn.
 - Tiết kiệm chi phí vận hành.

ILLUSTRATION 10.11

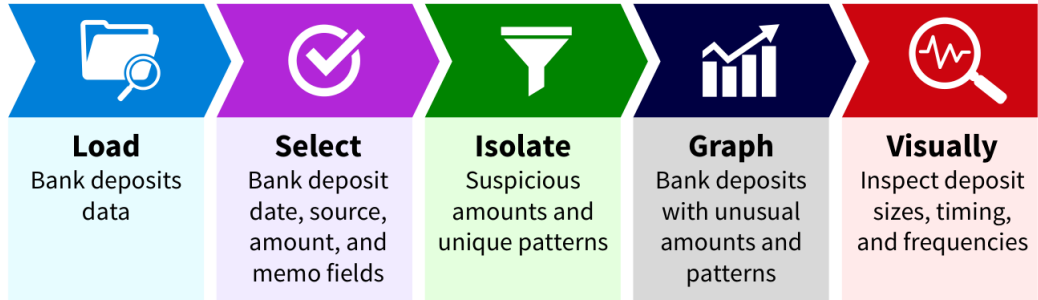


ILLUSTRATION 10.12

Traditional Accounting Ledgers			
Seller S's Records		Buyer B's Records	
↑ Cash	10	↑ Inventory	10
↑ Revenues	10	↓ Cash	10

Blockchain Shared Ledger			
Seller	Buyer	Date	Amount
S	B	9.14.22	10
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

Đạo đức Dữ liệu (Data Ethics)

- **Quyền riêng tư (Privacy):** Bảo mật thông tin nhận dạng cá nhân (PII) trong quá trình phân tích.
- **Tính minh bạch (Transparency):** Người dùng cần biết thuật toán AI đưa ra quyết định dựa trên điều gì.
- **Thiên lệch (Bias):** Dữ liệu lịch sử mang định kiến sẽ dạy cho AI đưa ra những quyết định phân biệt đối xử.

ILLUSTRATION 10.13

Agreement Type

Agreement Terms: performance and remittance responsibilities, consequences of contract breaches

Committed Assets: verification, fair market value, including any ownership transfers or bargain purchase options

Performance Schedule: timing, performance metrics

Scheduled Automatic Payments: timing, amounts, and automatic payment transfer information

Rights of each party

Secured: 

898DFA06F8D44E1..

Date of Agreement

Authenticated Agreement Parties

ILLUSTRATION 10.14



10.2 Công Nghệ Thay Đổi Phân Tích Dữ Liệu

- **Value Creation (Mục tiêu tạo giá trị):** Công nghệ không chỉ để lưu trữ mà phải tự động hóa, phân tích sâu và cung cấp dự đoán.
- Sự hội tụ của Cloud Computing, AI và Blockchain đang tái định hình cách bộ phận Kế toán làm việc hàng ngày.

ILLUSTRATION 10.15

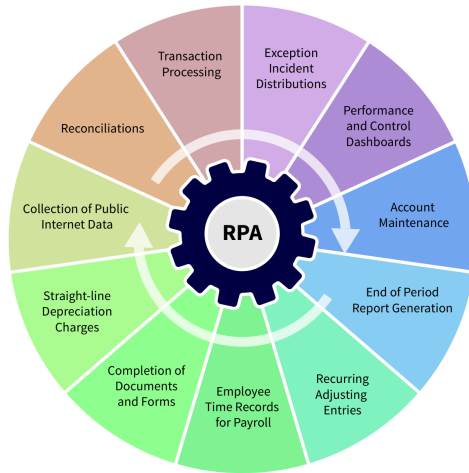
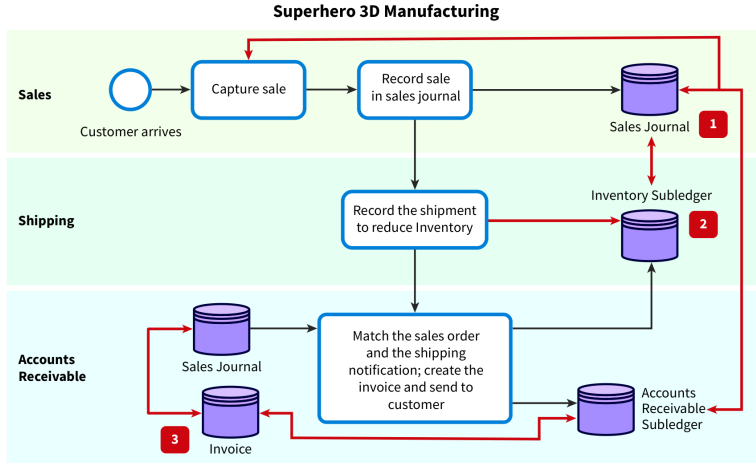


ILLUSTRATION 10.16



Khai phá Dữ liệu & Hợp đồng Thông minh

- **Khai phá Dữ liệu (Data Mining):** Ứng dụng thống kê và học máy để tìm kiếm mô hình (pattern) ẩn trong cơ sở dữ liệu lớn.
- **Hợp đồng Thông minh (Smart Contracts):** Code chạy trên blockchain tự động thực thi các điều khoản hợp đồng khi điều kiện được thỏa mãn (VD: tự động thanh toán khi hàng được giao).

ILLUSTRATION 10.17

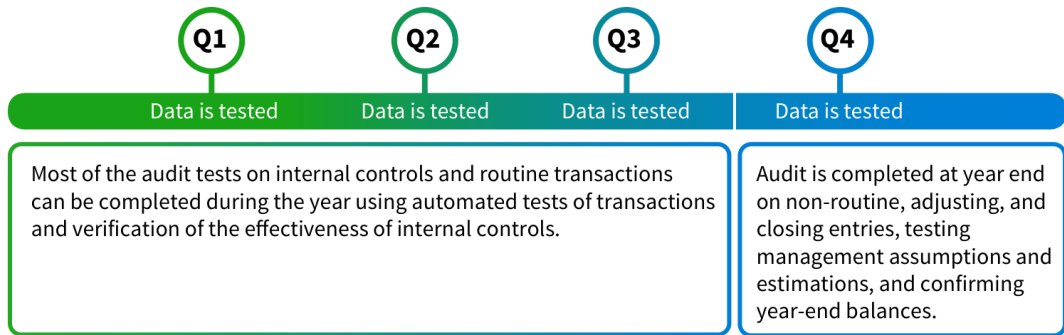


ILLUSTRATION 10.18

Optimistic Terms	Pessimistic Terms
	
Applaud	Bleak
Favorable	Costly
Successful	Troubled
Positive	Unstable

Tự Động Hóa Quy Trình Bằng Robot (RPA)

- **RPA (Robotic Process Automation):** Sử dụng phần mềm (bot) để tự động hóa các tác vụ lặp đi lặp lại có quy tắc cố định (VD: nhập liệu hóa đơn, đối chiếu ngân hàng).
- **Lợi ích:** Giảm thiểu lỗi sai (human error), giải phóng nhân sự kế toán cho công việc phân tích.

ILLUSTRATION 10.19

Corporate Sentiment Over Time

Updated: Nov 23, 2021 at 3:05 AM UTC

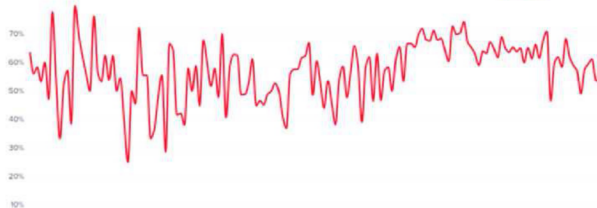
The chart below displays the Corporate Sentiment over time. We measure the current level of corporate leaders' "positivity," derived from English-language statements made by companies globally. The number displayed below shows the current sentiment out of a possible score of "100% positivity" in the range of the highest and lowest scores over the previous 365 days. Want to learn more? [Click here.](#)

Current Sentiment

53%
Positive

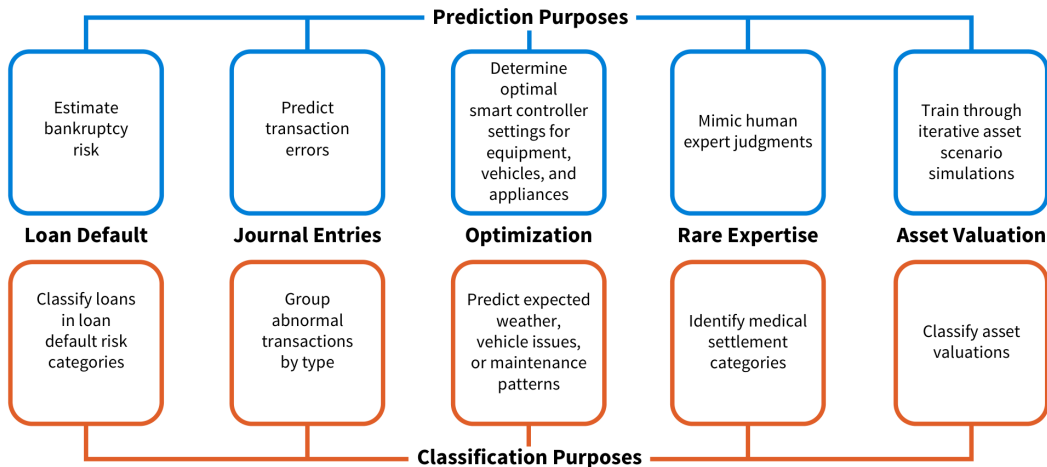
Ranging from 46% to
74% over the last year

Powered by  **Diligent**



Source: www.diligentinstitute.com/sentiment-tracker/

ILLUSTRATION 10.20



Khai Phá Quy Trình (Process Mining)

- **Process Mining:** Kỹ thuật phân tích nhật ký sự kiện (event logs) từ hệ thống (ERP) để phát hiện quy trình thực tế diễn ra như thế nào.
- **Ứng dụng:** Tìm ra các điểm nghẽn (bottleneck) trong quy trình phê duyệt tài chính, hoặc các bước thừa thãi gây lãng phí.

ILLUSTRATION 10.21

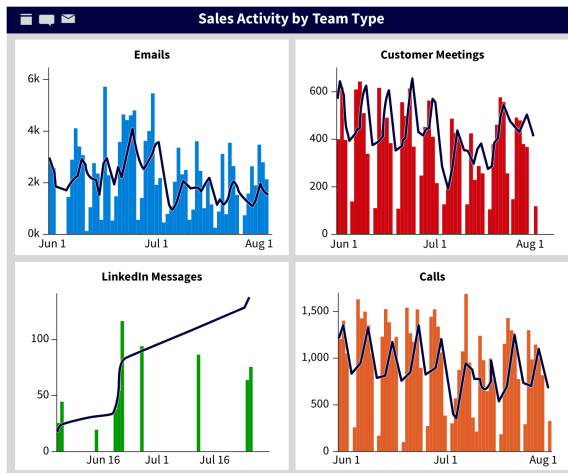


ILLUSTRATION 10.22



Kiểm Toán Liên Tục (Continuous Auditing)

- **Khái niệm:** Tự động hóa quá trình giám sát giao dịch tài chính liên tục theo thời gian thực.
- **Cách hoạt động:** Hệ thống cài đặt các ngưỡng báo động (alarms). Bất cứ giao dịch nào vi phạm rule (VD: giao dịch trong ngày nghỉ) sẽ bị đánh dấu ngay lập tức.

ILLUSTRATION 10.23

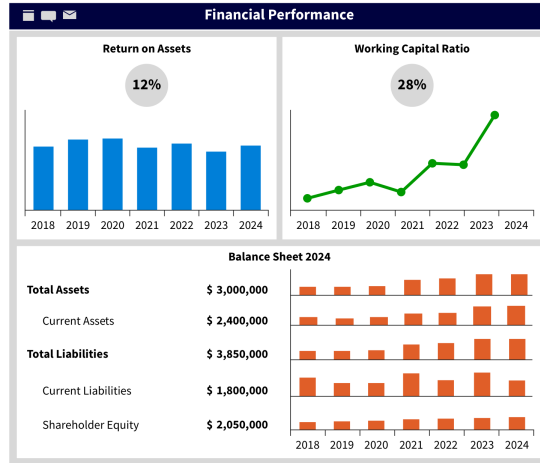
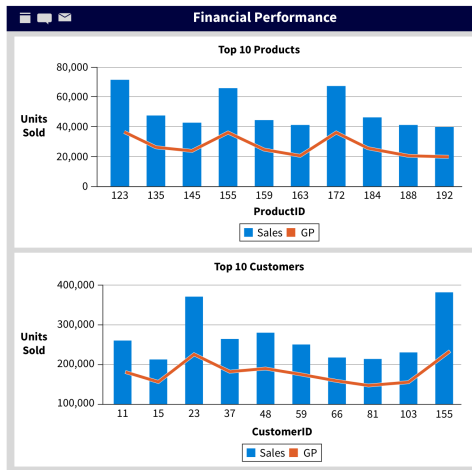


ILLUSTRATION 10.24



Phân Tích Văn Bản (Textual Analysis)

- **Textual Analysis:** Sử dụng NLP (Xử lý ngôn ngữ tự nhiên) để trích xuất thông tin từ tài liệu phi cấu trúc (Báo cáo thường niên 10-K, Biên bản họp).
- **Phân tích cảm xúc (Sentiment Analysis):** Đo lường tông giọng lạc quan hay bi quan của Ban giám đốc để dự báo hiệu quả tương lai.

ILLUSTRATION 10.25

Federal Tax Book Income vs. Taxable Income

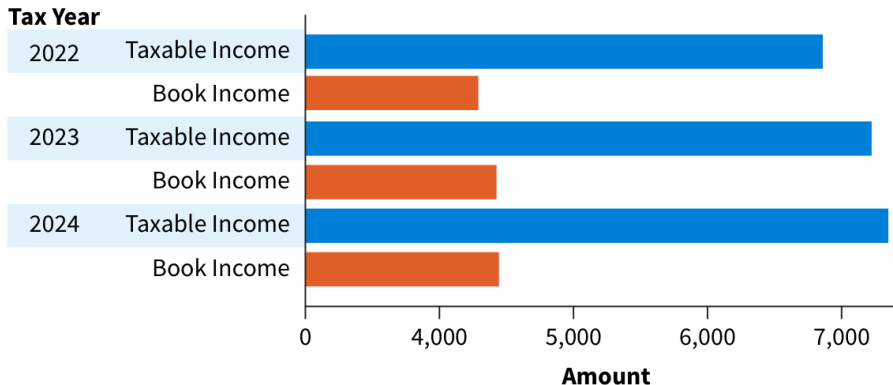


ILLUSTRATION 10.26

1

	A	B	C	D	E	F	G	H	I	J	K	L
1	Year	Month	Product	SumUnits Sold	Price	VCunit	Traceable FCunit	Common FCunit	Total Revenues	Total DirectVC	TotalApplied TraceableFC	TotalApplied CommonFC
2	2025	January	A	600	\$ 60.00	36.00	17.50	3.50	\$ 36,000.00	\$ 21,600.00	\$ 10,500.00	\$ 2,100.00
3	2025	January	B	1,710	\$ 100.00	80.00	18.75	3.50	\$ 171,000.00	\$ 136,800.00	\$ 32,062.50	\$ 5,985.00
4	2025	February	A	700	\$ 60.00	36.00	17.50	3.50	\$ 42,000.00	\$ 25,200.00	\$ 12,250.00	\$ 2,450.00
5	2025	February	B	1,640	\$ 100.00	80.00	18.75	3.50	\$ 164,000.00	\$ 131,200.00	\$ 30,750.00	\$ 5,740.00
6	2025	March	A	760	\$ 60.00	36.00	17.50	3.50	\$ 45,600.00	\$ 27,360.00	\$ 13,300.00	\$ 2,660.00
7	2025	March	B	1,360	\$ 100.00	80.00	18.75	3.50	\$ 136,000.00	\$ 108,800.00	\$ 25,500.00	\$ 4,760.00
8	2025	April	A	740	\$ 60.00	36.00	17.50	3.50	\$ 44,400.00	\$ 26,640.00	\$ 12,950.00	\$ 2,590.00
9	2025	April	B	1,270	\$ 100.00	80.00	18.75	3.50	\$ 127,000.00	\$ 101,600.00	\$ 23,812.50	\$ 4,445.00
10	2025	May	A	820	\$ 60.00	36.00	17.50	3.50	\$ 49,200.00	\$ 29,520.00	\$ 14,350.00	\$ 2,870.00
11	2025	May	B	1,200	\$ 100.00	80.00	18.75	3.50	\$ 120,000.00	\$ 96,000.00	\$ 22,500.00	\$ 4,200.00
12	2025	June	A	900	\$ 60.00	36.00	17.50	3.50	\$ 54,000.00	\$ 32,400.00	\$ 15,750.00	\$ 3,150.00
13	2025	June	B	940	\$ 100.00	80.00	18.75	3.50	\$ 94,000.00	\$ 75,200.00	\$ 17,625.00	\$ 3,290.00
14	2025	July	A	980	\$ 60.00	36.00	17.50	3.50	\$ 58,800.00	\$ 35,280.00	\$ 17,150.00	\$ 3,430.00
15	2025	July	B	860	\$ 100.00	80.00	18.75	3.50	\$ 86,000.00	\$ 68,800.00	\$ 16,125.00	\$ 3,010.00
16	2025	August	A	1,040	\$ 60.00	36.00	17.50	3.50	\$ 62,400.00	\$ 37,440.00	\$ 18,200.00	\$ 3,640.00
17	2025	August	B	720	\$ 100.00	80.00	18.75	3.50	\$ 72,000.00	\$ 57,600.00	\$ 13,500.00	\$ 2,520.00
18	2025	September	A	1,220	\$ 60.00	36.00	17.50	3.50	\$ 73,200.00	\$ 43,920.00	\$ 21,350.00	\$ 4,270.00
19	2025	September	B	650	\$ 100.00	80.00	18.75	3.50	\$ 65,000.00	\$ 52,000.00	\$ 12,187.50	\$ 2,275.00
20	2025	October	A	1,300	\$ 60.00	36.00	17.50	3.50	\$ 78,000.00	\$ 46,800.00	\$ 22,750.00	\$ 4,550.00
21	2025	October	B	600	\$ 100.00	80.00	18.75	3.50	\$ 60,000.00	\$ 48,000.00	\$ 11,250.00	\$ 2,100.00
22	2025	November	A	1,380	\$ 60.00	36.00	17.50	3.50	\$ 82,800.00	\$ 49,680.00	\$ 24,150.00	\$ 4,830.00
23	2025	November	B	550	\$ 100.00	80.00	18.75	3.50	\$ 55,000.00	\$ 44,000.00	\$ 10,312.50	\$ 1,925.00
24	2025	December	A	1,560	\$ 60.00	36.00	17.50	3.50	\$ 93,600.00	\$ 56,160.00	\$ 27,300.00	\$ 5,460.00
25	2025	December	B	500	\$ 100.00	80.00	18.75	3.50	\$ 50,000.00	\$ 40,000.00	\$ 9,375.00	\$ 1,750.00

Sheet1

Trí Tuệ Nhân Tạo (AI) & Công Nghệ Nhận Thức

- **Công nghệ nhận thức (Cognitive Technologies):** Giúp phần mềm có khả năng nhìn (Computer Vision), nghe, đọc và đưa ra quyết định như con người.
- **Machine Learning (Học máy):** Thuật toán có khả năng tự cải thiện theo thời gian mà không cần lập trình lại tường minh (VD: Nhận diện chứng từ giả).

ILLUSTRATION 10.27

	A	B	C	D	E	F	G	H	I	J	K	L
1	Year	Month	Product	SumUnits Sold	Price	VCunit	Traceable FCunit	Common FCunit	Total Revenues	Total Direct VC	Total Applied TraceableFC	TotalApplied CommonFC
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4	2025	March	A	760	\$ 60.00	36.00	17.50	3.50	\$ 45,600.00	\$ 27,360.00	\$ 13,300.00	\$ 2,660.00
5	2025	April	A	740	\$ 60.00	36.00	17.50	3.50	\$ 44,400.00	\$ 26,640.00	\$ 12,950.00	\$ 2,590.00
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8	2025	July	A	980	\$ 60.00	36.00	17.50	3.50	\$ 58,800.00	\$ 35,280.00	\$ 17,150.00	\$ 3,430.00
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20	2025	May	B	1,200	\$ 100.00	80.00	18.75	3.50	\$ 120,000.00	\$ 96,000.00	\$ 22,500.00	\$ 4,200.00
21	2025	June	B	940	\$ 100.00	80.00	18.75	3.50	\$ 94,000.00	\$ 75,200.00	\$ 17,625.00	\$ 3,290.00
22	2025	July	B	860	\$ 100.00	80.00	18.75	3.50	\$ 86,000.00	\$ 68,800.00	\$ 16,125.00	\$ 3,010.00
23	2025	August	B	720	\$ 100.00	80.00	18.75	3.50	\$ 72,000.00	\$ 57,600.00	\$ 13,500.00	\$ 2,520.00
24	2025	September	B	650	\$ 100.00	80.00	18.75	3.50	\$ 65,000.00	\$ 52,000.00	\$ 12,187.50	\$ 2,275.00
25	2025	October	B	600	\$ 100.00	80.00	18.75	3.50	\$ 60,000.00	\$ 48,000.00	\$ 11,250.00	\$ 2,100.00
26	2025	November	B	550	\$ 100.00	80.00	18.75	3.50	\$ 55,000.00	\$ 44,000.00	\$ 10,312.50	\$ 1,925.00
27	2025	December	B	500	\$ 100.00	80.00	18.75	3.50	\$ 50,000.00	\$ 40,000.00	\$ 9,375.00	\$ 1,750.00

ILLUSTRATION 10.28

	A	B	C	D	E	F	G	H	I	J	K	L
	Year	Month	Product	SumUnits Sold	Price	VCunit	Traceable FCunit	Common FCunit	Total Revenues	Total DirectVC	TotalApplied TraceableFC	TotalApplied CommonFC
1	2025	January	A	600	\$ 60.00	36.00	17.50	3.50	\$ 36,000.00	\$ 21,600.00	\$ 10,500.00	\$ 2,100.00
2	2025	February	A	700	\$ 60.00	36.00	17.50	3.50	\$ 42,000.00	\$ 25,200.00	\$ 12,250.00	\$ 2,450.00
3	2025	March	A	760	\$ 60.00	36.00	17.50	3.50	\$ 45,600.00	\$ 27,360.00	\$ 13,300.00	\$ 2,660.00
4	2025	April	A	740	\$ 60.00	36.00	17.50	3.50	\$ 44,400.00	\$ 26,640.00	\$ 12,950.00	\$ 2,590.00
5	2025	May	A	820	\$ 60.00	36.00	17.50	3.50	\$ 49,200.00	\$ 29,520.00	\$ 14,350.00	\$ 2,870.00
6	2025	June	A	900	\$ 60.00	36.00	17.50	3.50	\$ 54,000.00	\$ 32,400.00	\$ 15,750.00	\$ 3,150.00
7	2025	July	A	980	\$ 60.00	36.00	17.50	3.50	\$ 58,800.00	\$ 35,280.00	\$ 17,150.00	\$ 3,430.00
8	2025	August	A	1,040	\$ 60.00	36.00	17.50	3.50	\$ 62,400.00	\$ 37,440.00	\$ 18,200.00	\$ 3,640.00
9	2025	September	A	1,220	\$ 60.00	36.00	17.50	3.50	\$ 73,200.00	\$ 43,920.00	\$ 21,350.00	\$ 4,270.00
10	2025	October	A	1,300	\$ 60.00	36.00	17.50	3.50	\$ 78,000.00	\$ 46,800.00	\$ 22,750.00	\$ 4,550.00
11	2025	November	A	1,380	\$ 60.00	36.00	17.50	3.50	\$ 82,800.00	\$ 49,680.00	\$ 24,150.00	\$ 4,830.00
12	2025	December	A	1,560	\$ 60.00	36.00	17.50	3.50	\$ 93,600.00	\$ 56,160.00	\$ 27,300.00	\$ 5,460.00
13				12,000					\$ 720,000.00	\$ 432,000.00	\$ 210,000.00	\$ 42,000.00
14	2025	January	B	1,710	\$ 100.00	80.00	18.75	3.50	\$ 171,000.00	\$ 136,800.00	\$ 32,062.50	\$ 5,985.00
15	2025	February	B	1,640	\$ 100.00	80.00	18.75	3.50	\$ 164,000.00	\$ 131,200.00	\$ 30,750.00	\$ 5,740.00
16	2025	March	B	1,360	\$ 100.00	80.00	18.75	3.50	\$ 136,000.00	\$ 108,800.00	\$ 25,500.00	\$ 4,760.00
17	2025	April	B	1,270	\$ 100.00	80.00	18.75	3.50	\$ 127,000.00	\$ 101,600.00	\$ 23,812.50	\$ 4,445.00
18	2025	May	B	1,200	\$ 100.00	80.00	18.75	3.50	\$ 120,000.00	\$ 96,000.00	\$ 22,500.00	\$ 4,200.00
19	2025	June	B	940	\$ 100.00	80.00	18.75	3.50	\$ 94,000.00	\$ 75,200.00	\$ 17,625.00	\$ 3,290.00
20	2025	July	B	860	\$ 100.00	80.00	18.75	3.50	\$ 86,000.00	\$ 68,800.00	\$ 16,125.00	\$ 3,010.00
21	2025	August	B	720	\$ 100.00	80.00	18.75	3.50	\$ 72,000.00	\$ 57,600.00	\$ 13,500.00	\$ 2,520.00
22	2025	September	B	650	\$ 100.00	80.00	18.75	3.50	\$ 65,000.00	\$ 52,000.00	\$ 12,187.50	\$ 2,275.00
23	2025	October	B	600	\$ 100.00	80.00	18.75	3.50	\$ 60,000.00	\$ 48,000.00	\$ 11,250.00	\$ 2,100.00
24	2025	November	B	550	\$ 100.00	80.00	18.75	3.50	\$ 55,000.00	\$ 44,000.00	\$ 10,312.50	\$ 1,925.00
25	2025	December	B	500	\$ 100.00	80.00	18.75	3.50	\$ 50,000.00	\$ 40,000.00	\$ 9,375.00	\$ 1,750.00
26				12,000					\$ 1,200,000.00	\$ 960,000.00	\$ 225,000.00	\$ 42,000.00

10.3 Công Nghệ Gia Tăng Giá Trị Kế Toán

- Sự thay đổi diễn ra ở tất cả các lĩnh vực cốt lõi của ngành:
 - Hệ thống Thông tin Kế toán (AIS)
 - Kiểm toán (Auditing)
 - Kế toán Tài chính (Financial Accounting)
 - Kế toán Quản trị (Managerial Accounting)
 - Kế toán Thuế (Tax Accounting)

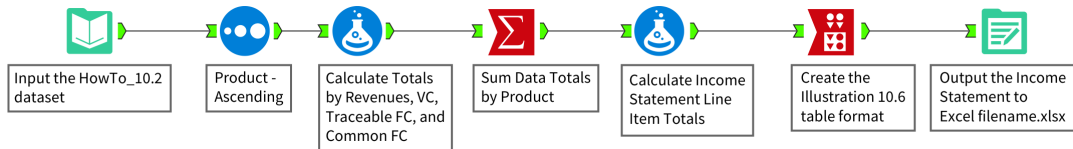
ILLUSTRATION 10.29

4 AutoSave Off

	N	O	P	Q	S
4	2025 Income Statement				
5		ProductA	ProductB	Both Product Lines	Only ProductA
6	Sales Revenues	+I14	+I28	=O6+P6	+O6
7	Less: Variable Costs	+J14	+J28	=O7+P7	+O7
8	Contribution Margin	=O6-O7	=P6-P7	=Q6-Q7	+O8
9	Less: Avoidable Fixed				
10	Expenses	+K14	+K28	+O10+P10	+O10
11	Segment Margin	=O8-O10	=P8-P10	=Q8-Q10	+O11
12	Less: Unavoidable Common				
13	Fixed Expenses	+L14	+L28	=O13+P13	=O13+P13
14	Net Income	=O11-O13	=P11-P13	=Q11-Q13	=S11-S13

Sheet1

ILLUSTRATION 10.30



Hệ thống Thông tin Kế toán (AIS)

- Hệ thống ERP hiện đại (như SAP, Oracle) đã tích hợp sẵn AI.
- **Vai trò của Kế toán:** Trở thành người giám sát, đảm bảo chất lượng dữ liệu đầu vào và thiết lập các rule chuẩn cho RPA tự động chạy.

ILLUSTRATION 10.31

The screenshot displays the Alteryx Designer interface. The top menu bar includes File, Edit, View, Options, and Help. Below it is a toolbar with various tool icons. The main workspace shows a workflow with two tools: 'Input the HowTo_10.2 dataset' and 'Product-Ascending'. A red circle highlights the 'Input Data' tool icon in the toolbar, and another red circle highlights the 'Sort' tool icon. A red arrow points from the 'Input Data' tool icon to the 'Input the HowTo_10.2 dataset' tool. Another red arrow points from the 'Sort' tool icon to the 'Product-Ascending' tool. A third red circle highlights the 'Run' button in the top right corner of the workflow canvas. The left sidebar shows the 'Sort (2) - Annotation' panel with the following details:

- Type: Sort
- ID: 2
- Name: Sort (2)
- Show Annotation: Using Canvas Settings
- Place Annotation on the Top: ☐
- Annotation: Product - Ascending

The bottom section of the interface shows the 'Results - Sort (2) - Output' table. It indicates 12 of 12 fields are visible and 24 records are displayed. The table has columns for Record, Year, Month, Product, SumUnits Sold, Price, VCunit, Traceable FCunit, Common FCunit, and Total Revenues. The first record shows data for 2025, January, Product A, with 600 units sold at a price of 60, resulting in 36 VCunits, 17.5 Traceable FCunits, 3.5 Common FCunits, and 36000 Total Revenues.

Record	Year	Month	Product	SumUnits Sold	Price	VCunit	Traceable FCunit	Common FCunit	Total Revenues
1	2025	January	A	600	60	36	17.5	3.5	36000

ILLUSTRATION 10.32

The screenshot displays the Alteryx Designer interface. The top menu bar includes File, Edit, View, Options, and Help. Below it is a toolbar with various tools like Favorites, Recommended, In/Out, Preparation, Join, Parse, Transform, In-Database, Reporting, and Documentation. The main workspace is divided into three panes:

- Formula (3) - Configuration:** This pane shows the configuration for the Formula tool. It includes a list of output columns and their data types. The first column is 'RevenuesA' with a value of 36000. The second column is 'RevenuesB' with a value of 0. The third column is 'VCforA' with a value of 21600. The fourth column is 'VCforB' with a value of 0. The fifth column is 'TraceFCforA' with a value of 10500. The sixth column is 'TraceFCforB' with a value of 0. The seventh column is 'CommonFCforA' with a value of 2100. The eighth column is 'CommonFCforB' with a value of 0. The data type for all columns is set to 'Double' with a size of 8.
- How To 10.2 Workflow.xmd*:** This pane shows a workflow diagram. It includes a red circle around the 'Create Calculated Field Columns' tool, which is connected to the 'Product-Ascending' tool. A red line with the number '4' points to the 'Formula' tool in the top toolbar.
- Results - Formula (3) - Input:** This pane shows a data preview table with 13 records and 7 columns: Record, SumUnits Sold, Price, VCunit, Traceable FCunit, Common FCunit, and Total Revenues.

Record	SumUnits Sold	Price	VCunit	Traceable FCunit	Common FCunit	Total Revenues
1	600	60	36	17.5	3.5	36000
2	700	60	36	17.5	3.5	42000
3	760	60	36	17.5	3.5	45600
4	740	60	36	17.5	3.5	44400
5	820	60	36	17.5	3.5	49200
6	900	60	36	17.5	3.5	54000
7	980	60	36	17.5	3.5	58800
8	1040	60	36	17.5	3.5	62400
9	1220	60	36	17.5	3.5	73200
10	1300	60	36	17.5	3.5	78000
11	1380	60	36	17.5	3.5	82800
12	1560	60	36	17.5	3.5	93600
13	1710	100	80	18.75	3.5	171000

Kiểm toán (Auditing)

- Không còn kiểm tra chọn mẫu. Phân tích dữ liệu cho phép kiểm tra 100% tổng thể dữ liệu giao dịch.
- Nhận diện rủi ro chủ động bằng học máy thay vì phản ứng thụ động sau khi kỳ kế toán kết thúc.

ILLUSTRATION 10.33

Summarize (4) - Configuration

Fields:

Field	Type
Year	Double
Month	V_String
Product	V_String
SumUnitsSold	Double
Price	Double
VCunit	Double
TraceableFCunit	Double
CommonFCunit	Double
TotalRevenues	Double
TotalDirectVC	Double
TotalAppliedTraceableFC	Double
TotalAppliedCommonFC	Double
RevenueA	Double
RevenueB	Double

Actions:

Field	Action	OutputFieldName
TotalRevenues	Sum	Sum_TotalRevenues
RevenuesA	Sum	Sum_RevenuesA
RevenuesB	Sum	Sum_RevenuesB
VCforA	Sum	Sum_VCforA
VCforB	Sum	Sum_VCforB
TraceFCforA	Sum	Sum_TraceFCforA
TraceFCforB	Sum	Sum_TraceFCforB
CommonFCforA	Sum	Sum_CommonFCforA
CommonFCforB	Sum	Sum_CommonFCforB

How To 10.2 Workflow.xmld*

Input the HowTo... 10.2 dataset → Product-Ascending → Create Calculated Field Columns → Create the totals needed for the Income Statement

Results - Summarize (4) - Input

20 of 20 Fields ▾ | 24 records displayed

Record	SumUnits Sold	Price	VCunit	Traceable FCunit	Common FCunit	Total Revenues
1	600	60	36	17.5	3.5	36000
2	700	60	36	17.5	3.5	42000
3	760	60	36	17.5	3.5	45600
4	740	60	36	17.5	3.5	44400
5	820	60	36	17.5	3.5	49200
6	900	60	36	17.5	3.5	54000
7	980	60	36	17.5	3.5	58800
8	1040	60	36	17.5	3.5	62400
9	1220	60	36	17.5	3.5	73200
10	1300	60	36	17.5	3.5	78000
11	1380	60	36	17.5	3.5	82800
12	1560	60	36	17.5	3.5	93600
13	1710	100	80	18.75	3.5	171000
14	1640	100	80	18.75	3.5	164000
15	1360	100	80	18.75	3.5	136000
16	1270	100	80	18.75	3.5	127000
17	1200	100	80	18.75	3.5	120000

ILLUSTRATION 10.34

The screenshot displays the Alteryx Designer interface. The top toolbar includes various tools, with the 'Formula' tool highlighted by a red circle and a red line connecting it to the 'Calculate more totals needed for the income statement' tool in the workflow. The 'Formula (5) - Configuration' window is open on the left, showing the configuration for three output columns: ContribMarginA, ContribMarginB, and SegmentMarginA. Each column has a formula, data type set to Double, and size set to 8. The main workflow area shows a sequence of tools: 'Input the HowTo_10.2 dataset', 'Product - Ascending', 'Create Calculated Field Columns', 'Create the totals needed for the income statement', and 'Calculate more totals needed for the income statement'. The 'Calculate more totals needed for the income statement' tool is highlighted with a red circle and a red number 6. Below the workflow, the 'Results - Formula (5) - Input' section shows a table with 9 fields and 1 record displayed.

Record	Sum_Total Revenues	Sum_ RevenuesA	Sum_ RevenuesB	Sum_ VCforA	Sum_ VCforB	Sum_Tra ceFCforA
1	1920000	720000	1200000	432000	960000	210000

- Phân tích văn bản trên hàng ngàn BCTC 10-K của đối thủ trong vài giây để đưa ra báo cáo so sánh.
- Hỗ trợ lập BCTC chuẩn XBRL tự động và phát hiện bất thường trước khi nộp lên Ủy ban Chứng khoán.

ILLUSTRATION 10.35A

Output Column Header	Description Field	Fields to Select
Product A	Add New Description	SumRevenuesA
		SumVCforA
		ContribMarginA
		SumTraceFCforA
		SegmentMarginA
		SumCommonFCforA
		NetIncomeA
Product B	None	SumRevenuesB
		SumVCforB
		ContribMarginB
		SumTraceFCforB
		SegmentMarginB
		SumCommonFCforB
		NetIncomeB
Both Products	None	SumTotalRevenues
		TotalVC
		TotalContribMargin
		TotalTraceFC
		TotalSegmentMargin
		TotalCommonFC
		TotalNetIncome

(Continued)

- Thay vì chỉ nhìn vào quá khứ (Descriptive), Kế toán Quản trị chuyển sang Dự báo (Predictive) và Đề xuất (Prescriptive).
- Ứng dụng Process Mining để tối ưu hóa chi phí vận hành chuỗi cung ứng.

ILLUSTRATION 10.35B

Output Column Header	Description Field	Fields to Select
Only Product A	None	SumRevenuesA
		SumVCforA
		ContribMarginA
		SumTraceFCforA
		SegmentMarginA
		TotalCommonFC
		NetIncomeOnlyA

- RPA tự động điền tờ khai và đối chiếu quy định pháp luật thay đổi liên tục.
- Phân tích rủi ro hồ sơ bị thanh tra thuế để doanh nghiệp có phương án giải trình sớm.

ILLUSTRATION 10.36

The screenshot displays the Alteryx Designer software interface. The top menu bar includes File, Edit, View, Options, and Help. Below it, a toolbar shows various tool categories: Favorites, Recommended, In/Out, Preparation, Join, Parse, Transform, Machine Learning, Text Mining, and Computer Vision. The 'Arrange' tool icon is circled in red in the toolbar. The main workspace is divided into three panels:

- Left Panel (Arrange (6) - Configuration):** Contains 'Key Fields' and 'Output Fields' sections. The 'Key Fields' section has a list of fields with checkboxes: Sum_TotalRevenues, Sum_RevenuesA, Sum_RevenuesB, and Sum_VCforA. The 'Output Fields' section shows a table with columns 'Description' and 'Product A'.
- Right Panel (New Workflow1*):** Displays a workflow diagram with several tools connected in sequence. The 'Arrange' tool is circled in red and labeled with a red '7'.
- Bottom Panel (Add Column):** A dialog box for adding a new column. It has a 'Column Header' field set to 'Product B' and a 'Fill in Description Column' dropdown set to '[None]'. Below, a list of fields is shown with checkboxes: Sum_RevenuesB, Sum_VCforB, Sum_TraceFCforB, Sum_CommonFCforB, ContribMarginB, SegmentMarginB, and NetIncomeB. The 'OK' button is highlighted.

Description	Product A
Revenues	Sum_RevenuesA
Variable Costs	Sum_VCforA
Contribution Margin	ContribMarginA
Traceable Fixed Costs	Sum_TraceFCforA
Segment Margin	SegmentMarginA
Common Fixed Costs	Sum_CommonFCforA
Net Income	NetIncomeA

Tổng Kết Chương 10

- Nghề kế toán không biến mất, mà sẽ chuyển dịch lên cấp độ cao hơn.
- Công việc chân tay, lặp lại sẽ do RPA và AI đảm nhiệm.
- Kế toán viên cần trở thành những **Data Analyst** và **Strategic Advisor** thực thụ.

ILLUSTRATION 10.37

The screenshot displays the Alteryx Designer interface. The top menu bar includes File, Edit, View, Options, and Help. Below it is a toolbar with various tools: Favorites, Recommended, In/Out, Preparation, Join, Parse, Transform, In-Database, Reporting, and Documentation. The 'Arrange' tool is highlighted in the toolbar and in the workflow diagram.

The 'Arrange (7) - Configuration' window is open on the left. It shows the 'Key Fields' section with a list of fields: Sum_TotalRevenues, Sum_RevenuesA, Sum_RevenuesB, and Sum_VCforA. The 'Output Fields' section shows a table with columns: Description, Product A and B, and Both Products. The table contains data for Sales Revenues, Variable Costs, Contribution Margin, Traceable Fixed Costs, Segment Margin, and Net Income.

The workflow diagram on the right shows a sequence of tools: Input the HowTo_10.2 dataset, Product - Ascending, Create Calculated Field Columns, Create the totals needed for the income statement, Calculate more totals needed for the income statement, and Create the formatted income statement. The 'Arrange' tool is the final step in the workflow.

Below the workflow diagram, the 'Results - Arrange (7) - Input' window shows a table with 22 fields. The first record is displayed:

Record	Sum_Total Revenues	Sum_RevenuesA	Sum_RevenuesB	Sum_VC forA	Sum_VC forB
1	1920000	720000	1200000	432000	960000

Apply It 10.1

Data Field	Data Characteristic
Sleeping bag fabric durability rating from quality testing.	
Sales revenues from repeat customers and new customers during the last 12 months.	
Cell phone data captured from bicycle rides using a navigation app.	
Variability of manufacturing equipment performance every five minutes during use.	
Naturally occurring bundles of camping equipment and outdoor clothing sales each month.	
Hourly outside temperatures during weekend days when most customers are using the equipment.	
Customer comments about their outdoor product satisfaction on social media platforms.	

SOLUTION

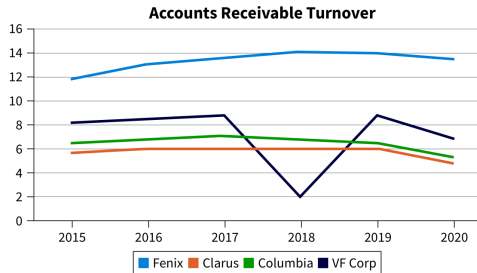
Data Field	Data Characteristic
Sleeping bag fabric durability rating from quality testing.	Value
Sales revenues from repeat customers compared to new customers during the last 12 months.	Veracity and value
Cell phone data captured from bicycle rides using a navigation app.	Volume and variety
Variability of manufacturing equipment performance every five minutes during use.	Volume
Naturally occurring bundles of camping equipment and outdoor clothing sales each month.	Value
Hourly outside temperatures during weekend days when most customers are using the equipment.	Volume
Customer comments about their outdoor product satisfaction on social media platforms.	Variety and volume

(This is a suggested solution; other relevant insights might also be correct.)

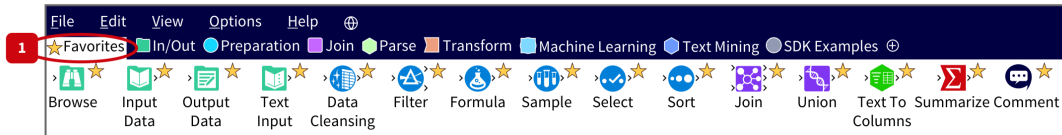
Apply It 10.2

Ratio Values and Graph from the Comparison worksheet.

Comparison of Accounts Receivable Turnovers						
	2015	2016	2017	2018	2019	2020
Fenix	11.9	13.1	13.6	14.1	14.0	13.5
Clarus	5.7	5.9	5.8	5.9	5.9	4.9
Columbia	6.5	6.7	7.1	6.9	6.5	5.3
VF Corp	8.3	8.6	8.8	2.1	8.7	6.9



HowTo 10.2



Adventure Sports 10.13

VendorMasterFile	PurchaseOrder	ReceivingReport	Invoices
VendorID VendorName VendorAddress VendorCity VendorState VendorZip VendorPayterms	PONo VendorID VendorName PODate POItemID POItemDescription POItemCost POItemQty POExpectedPreceiptDate	RECNO PONo VendorID VendorName ItemID ItemDescription RecItemQty RecDate	InvoiceNO PONo VendorID VendorName ItemID ItemDescription InvoiceQty InvoiceItemCost InvoiceTotal

Phần Bài tập Ngắn

Brief Exercises (BE)



Phần Bài tập

Exercises (EX)

EX 10.3 & EX 10.12

March 15, 20XX

Memo for the Audit Files

RE: Continuous Audit of Revenue Data & Professional Due Care Considerations

This memo documents the engagement team's procedures to provide reasonable assurance regarding the client's revenue-cycle transactions and the engagement team's procedures to ensure that the client's data is kept confidential. First, we discuss the audit procedures and second, we discuss professional due care considerations.

Continuous Audit of Revenue Procedures

The client has provided us access to all sales transaction data in real-time by allowing the audit firm to connect its proprietary analytics software to the company's relational database. As revenue transactions are recorded in the client's database, the auditor's software automatically tests and captures the audit data.

We performed audit procedures on this data throughout the year, including the following descriptive statistics and diagnostic tests:

1. _____

Based on these descriptive and diagnostic tests, and comparing our results to past years, abnormal transactions or anomaly transactions were identified and further investigated by collecting and reviewing their related source documents, such as contracts, invoices, and customer payments.

Professional Due Care Considerations

Since sales and collection transactions have a high data volume of personally identifiable data, great care must be taken to ensure that the client's data is kept private.

For example, 2. _____

F/S	AcctNo	AcctDescription	6/30/2025	6/30/2024	Difference	%Change
Income Statement		Warranty				
Account	32153	Expense	\$ 347,954	\$ 535,849	\$ (187,895)	-35%
Balance Sheet		Warranty				
Account	23829	Liability	\$ 121,784	\$ 241,132	\$ (119,348)	-49%

Hỏi & Đáp

Cảm ơn các bạn đã lắng nghe!